MEMORANDUM

TO: Mr. S. W. Dunwell

January 31st 1956

Project Stretch Requirements of the RAM
Supplement 1

Reference: Mr. S.W.Dunwell
Memo of 12-16-55

The following is a set of tentative specifications for the RAM for use on the Stretch Program. These have been arrived at through discussions with Messrs. Critchlow and Haug of San Jose, with various members of Applied Science and with the Stretch Planning Activity.

Tentative Specifications

1. Number of bits in parallel - 60 minimum with no overlap between adjacent groups of 60 bits.
2. Mode of recording - NRZ
3. Disc size - 24 inches
4. RPM - 1200 minimum
5. Bit rate - 6 microseconds per bit maximum
6. Storage capacity per 60 sides - 1,048,576 words
7. Bit density - 200 bits per inch
8. Number of tracks per side - 128
9. Number of bits per track - 8192
10. Number of blocks per track - 4
11. Average bit rate - 6 microseconds per bit
12. Instantaneous bit rate - 5.7 microseconds per bit

The following considerations were used in arriving at the above specifications:

Most important consideration

1. Maximum storage capacity
2. High bit rate (The bit rate must be 6 microseconds per bit or less)
3. The RAM must be addressed in the binary system

Secondary considerations

4. The revolution time should be kept as low as consistent with considerations 1 and 2 above.

5. The random access time from track to track should be kept low. This, however, is the least important of the considerations.

With the above specifications, the outer track diameter is only 21 inches. If it would be feasible to use a $21\frac{1}{2}$ or 22 inch disc and achieve a higher rpm, this should be considered. Due to the binary nature of the addressing scheme, if an increase in capacity is contemplated, it will be desirable to increase capacity by a full factor of 2. However, the rpm can be increased by any amount and should be as fast as possible consistent with the specifications.

The number of parallel bits in a word has been fixed at 60. However, this does not include the additional checking bits which will be required. The number of these will be between 1 and 8 depending upon the degree of checking and auto correction to be included.

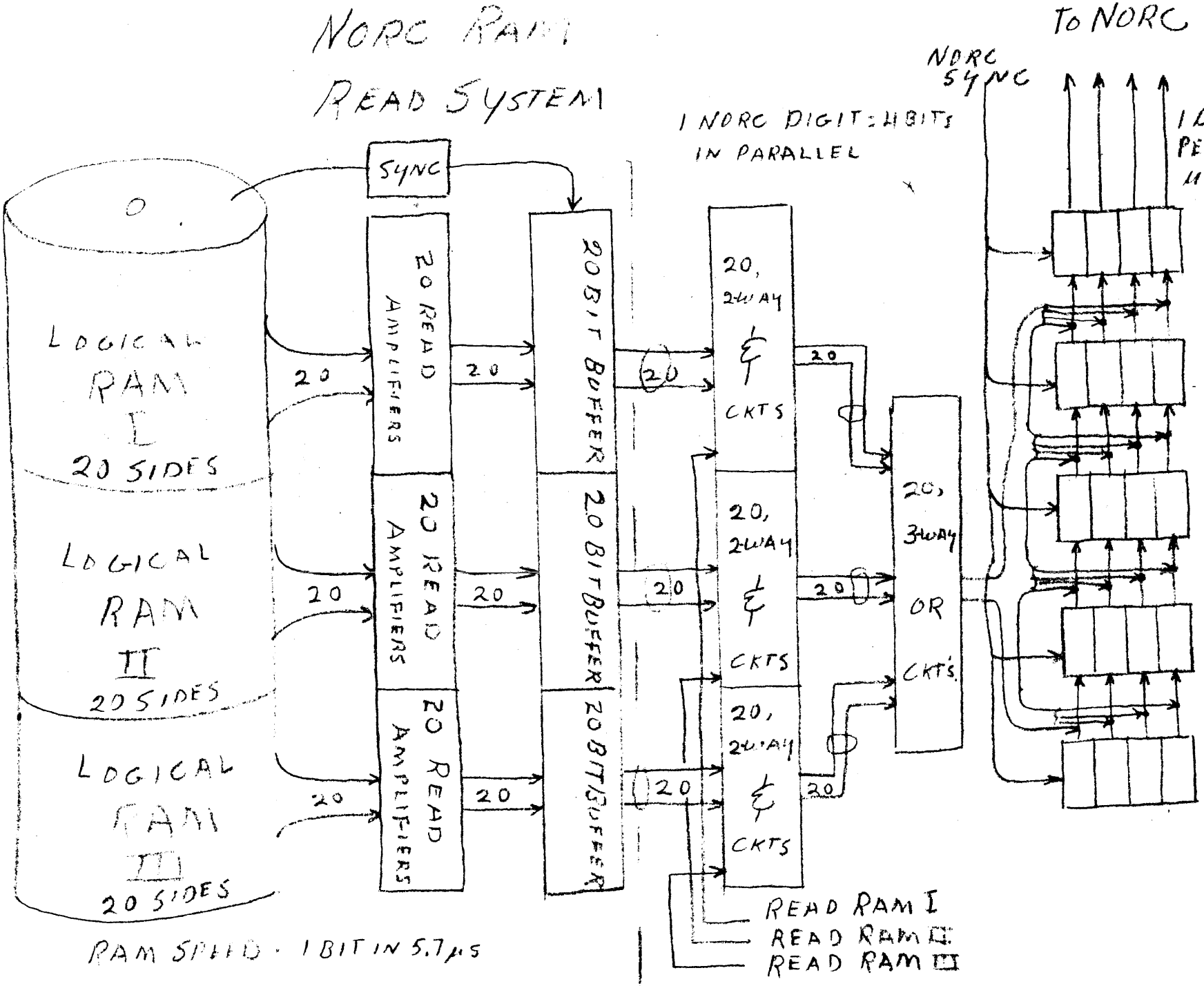
NPE

N. P. Edwards

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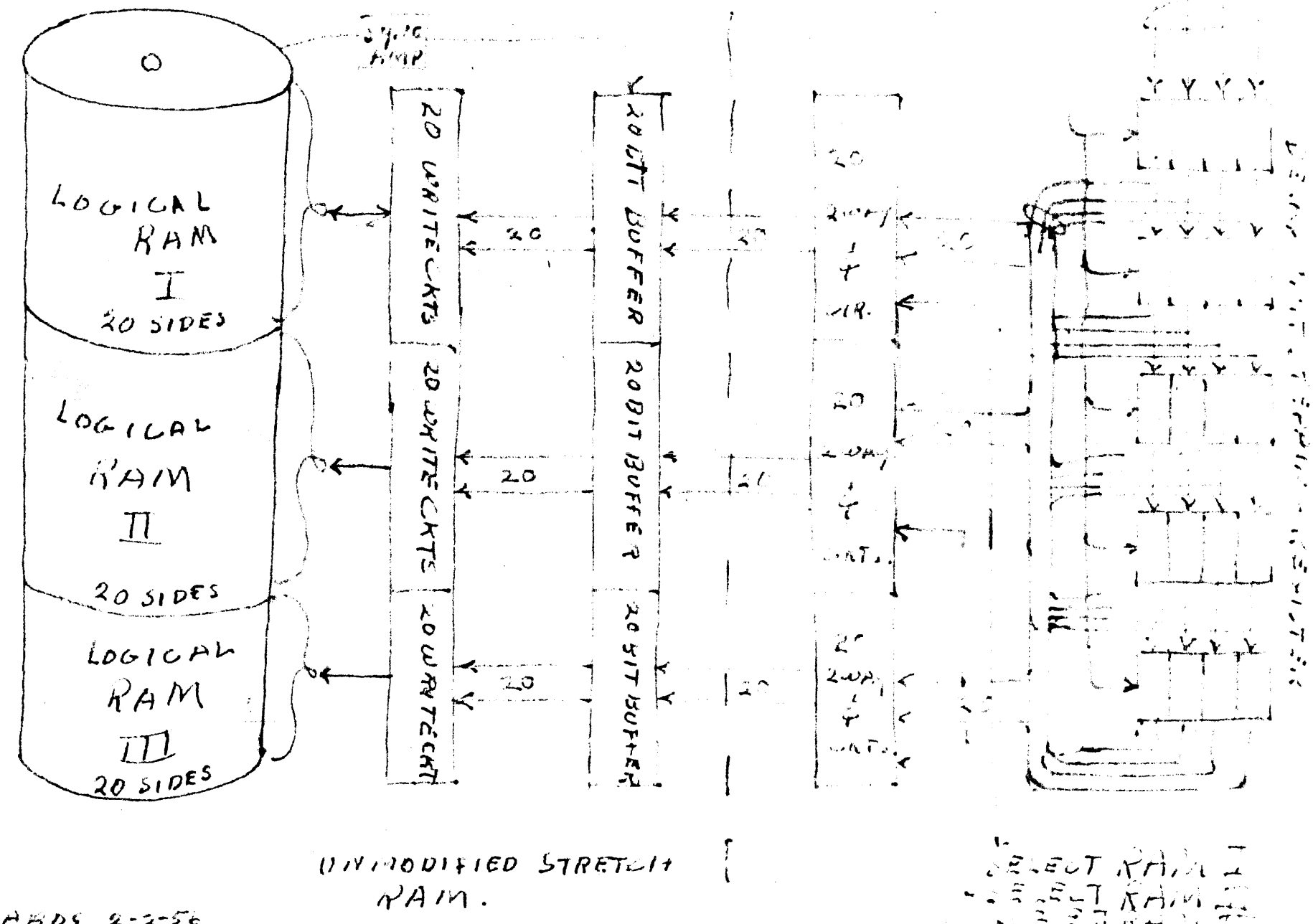
c.c. E. M. Boehm
J. E. Griffith
L. S. Snyder
H. Wild
W. Wolensky
J. Backus
A. J. Critchlow
R. L. Haug
R. Johnson
J. A. Haddad
B. O. Evans
M. E. Femmer
W. Buslik
W. G. Bouricius
N. Rochester
W. A. Hunt

NORC RAM READ SYSTEM



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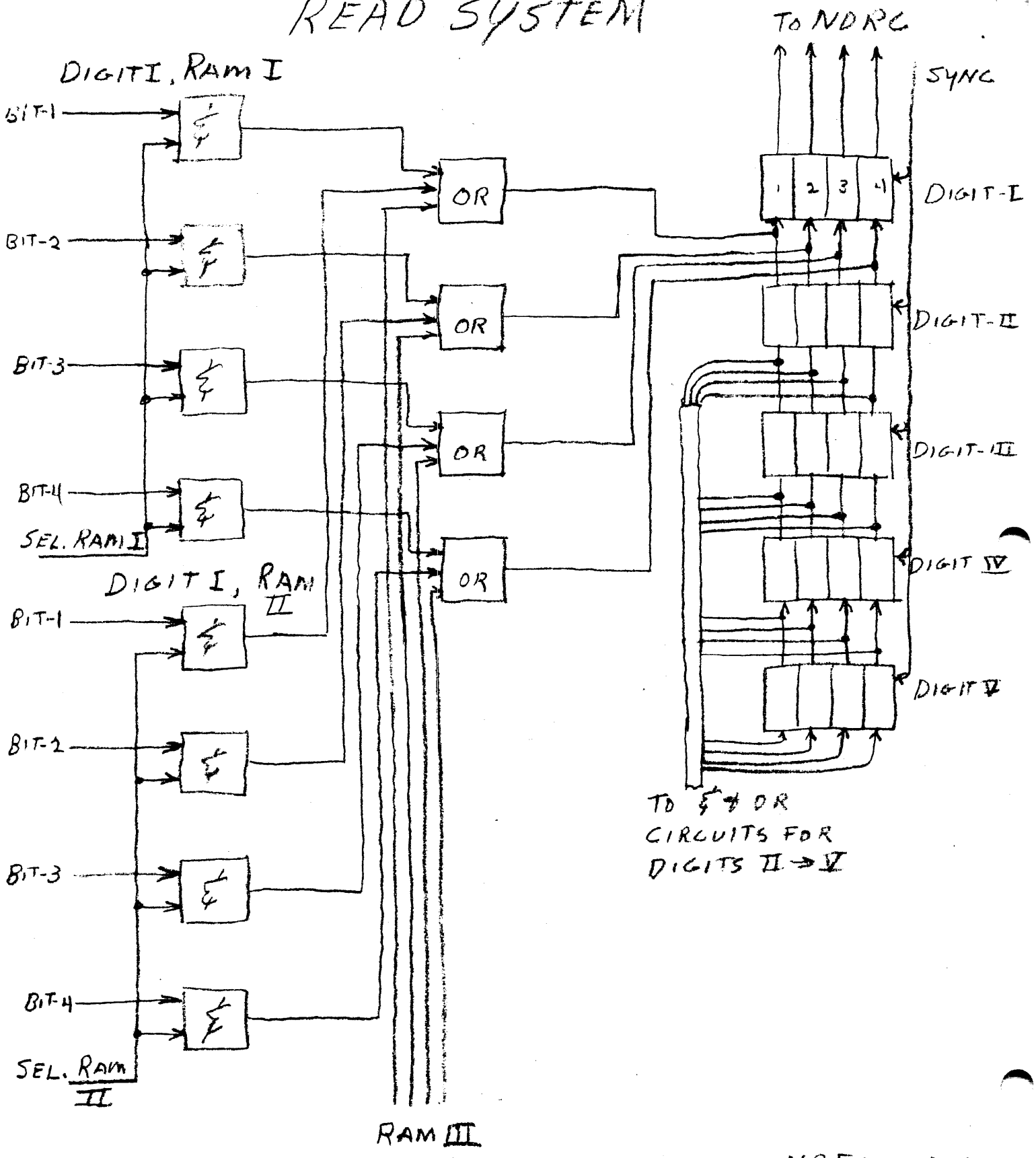
NORC RAM WRITE SYSTEM



UNMODIFIED STRETCH RAM.

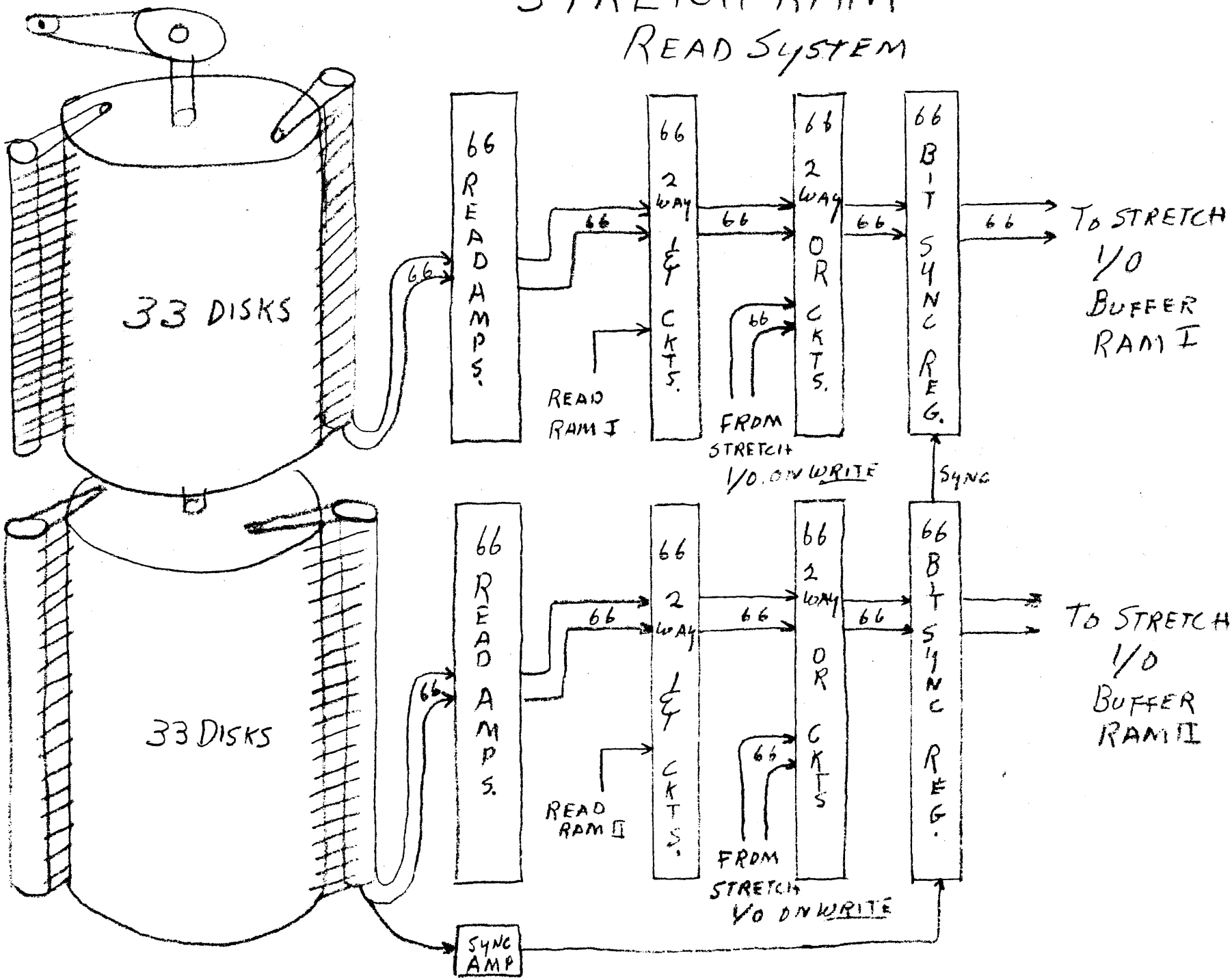
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NORC RAM READ SYSTEM



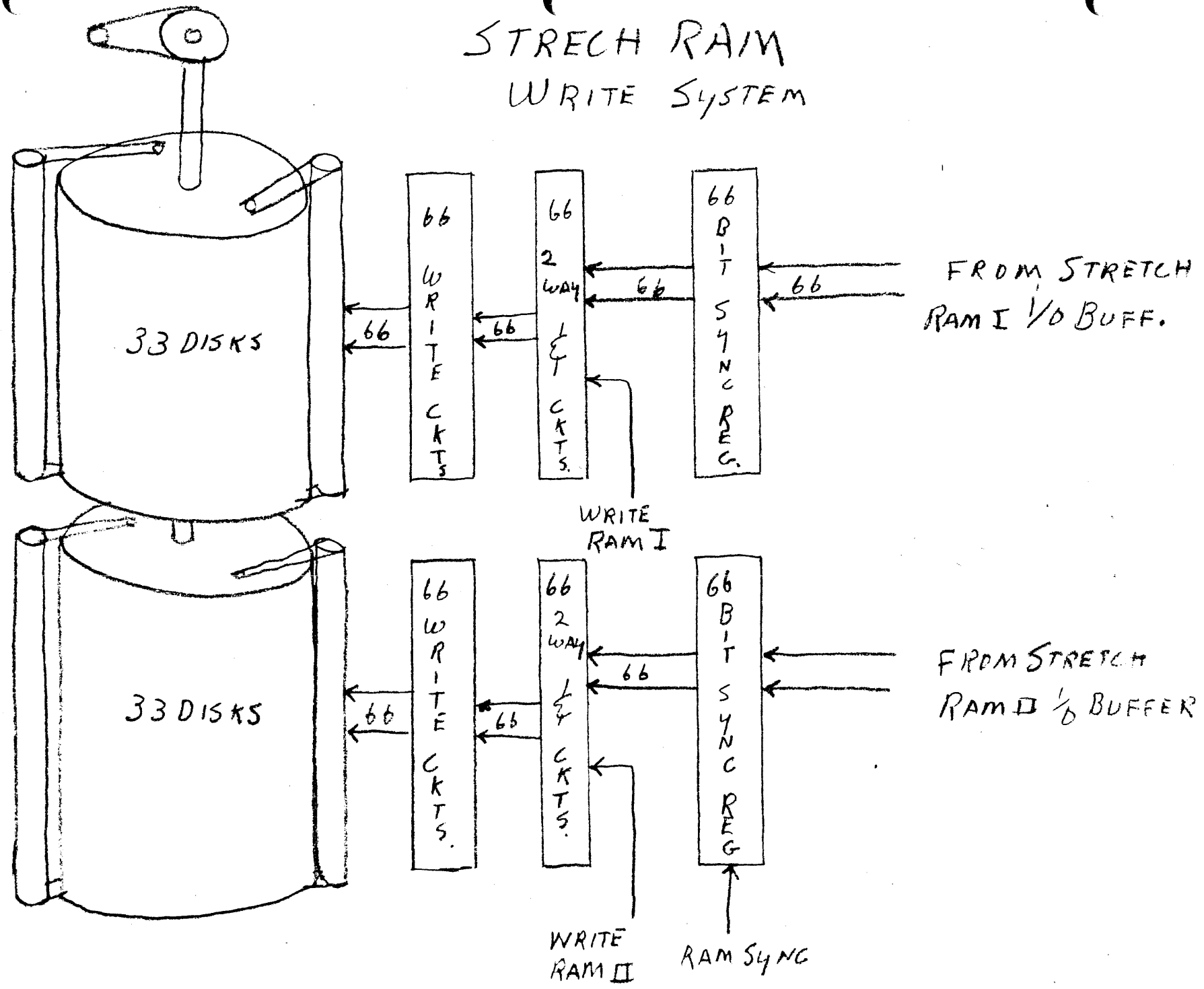
N.P. EDWARDS
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STRETCH RAM
READ SYSTEM



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STRETCH RAM
WRITE SYSTEM



M.P. EDWARDS 2-3-56